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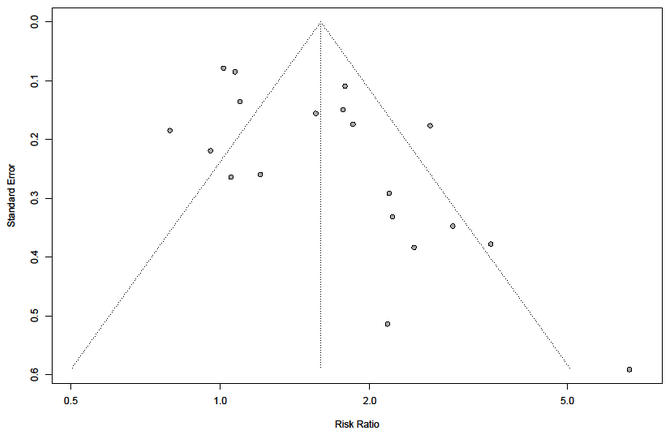
Supporting Information 1: full search strategy

1.digital therapy
#1 digital*[Title/Abstract]
#2 smartphone[Title/Abstract]
#3 smart-phone[Title/Abstract]
#4 computer*[Title/Abstract]
#5 mobile*[Title/Abstract]
#6 cellphone[Title/Abstract]
#7 app*[Title/Abstract]
#8 “mobile health” [Title/Abstract]
#9 telemedicine[Mesh Terms]
#10=#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9

2.tobacco use disorder
#1 tobacco use disorder[Mesh Terms]
#2 nicotine addiction[Title/Abstract]
#3 tobacco dependence[Title/Abstract]
#4 smoking cessation[Mesh Terms]
#5=#1 OR #2 OR #3 OR #4

Results:
Pubmed:1147
Cochrane:2849
Embase:2601
CNKI:17

Supporting Information 2: funnel plot

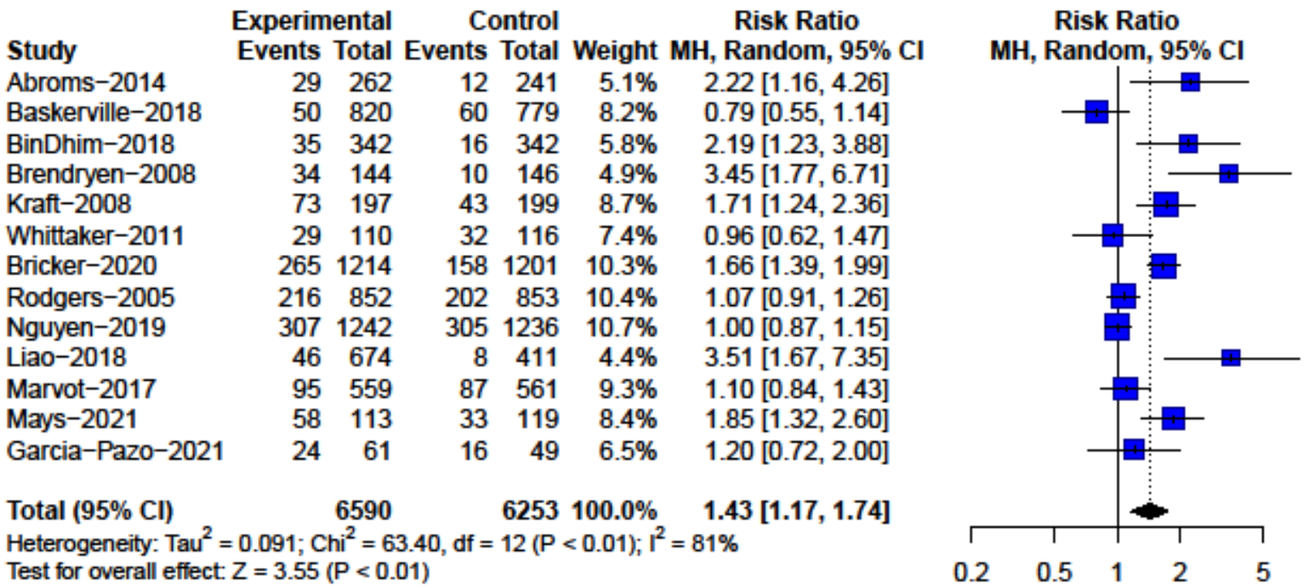


p-value = 0.1318

Supporting Information 3: Cochrane ROB2 summary table

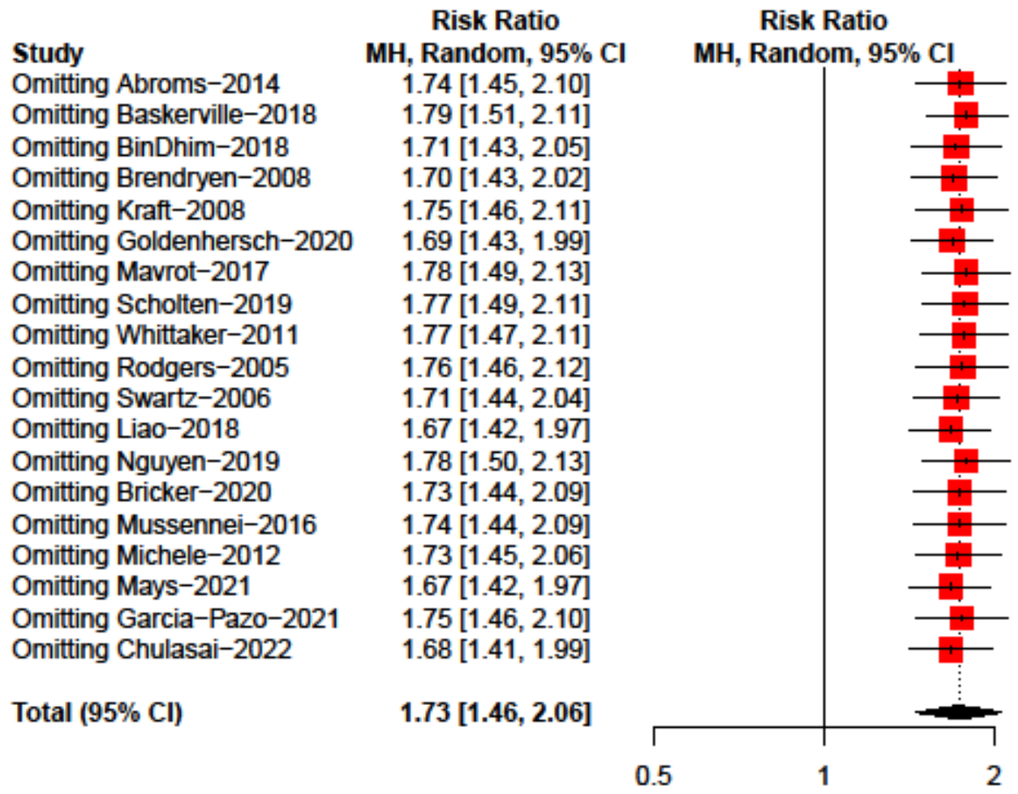
	Randomization process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Bias
Abroms 2014	Low	Low	Low	Low	Low	Low
Baskerville 2018	Low	Low	High	Low	Low	High
BinDhim 2018	Low	Low	Low	Low	Low	Low
Brendryen 2008	Low	Low	Low	Low	Low	Low
Kraft 2008	Low	Low	Low	Low	Low	Low
Goldenhersc 2020	Low	Low	High	Low	Low	High
Mavrot 2017	Low	Low	High	Low	Low	Some concerns
Scholten 2019	Low	Low	Low	Low	Low	Low
Whittaker 2011	Low	Low	Some concerns	Low	Low	Some concerns
Rodgers2005	Low	Low	Low	Low	Low	Low
Swartz 2006	Low	Low	High	Low	Low	High
Liao 2018	Low	Low	Low	Low	Low	Low
Nguyen 2019	Low	Low	High	Low	Low	High
Bricker 2020	Low	Low	Low	Low	Low	Low
Mussennei 2016	Low	Low	Low	Low	Low	Low
Michele 2012	Low	Low	High	Low	Low	High
Mays 2021	Low	Low	Low	Low	Low	Low
Garcia-Pazo 2021	Low	Low	Some concerns	Low	Low	Some concerns
Chulasai 2022	Low	Low	Low	Low	Low	Low

Supporting Information 4: forest plot of abstinence at 6-month

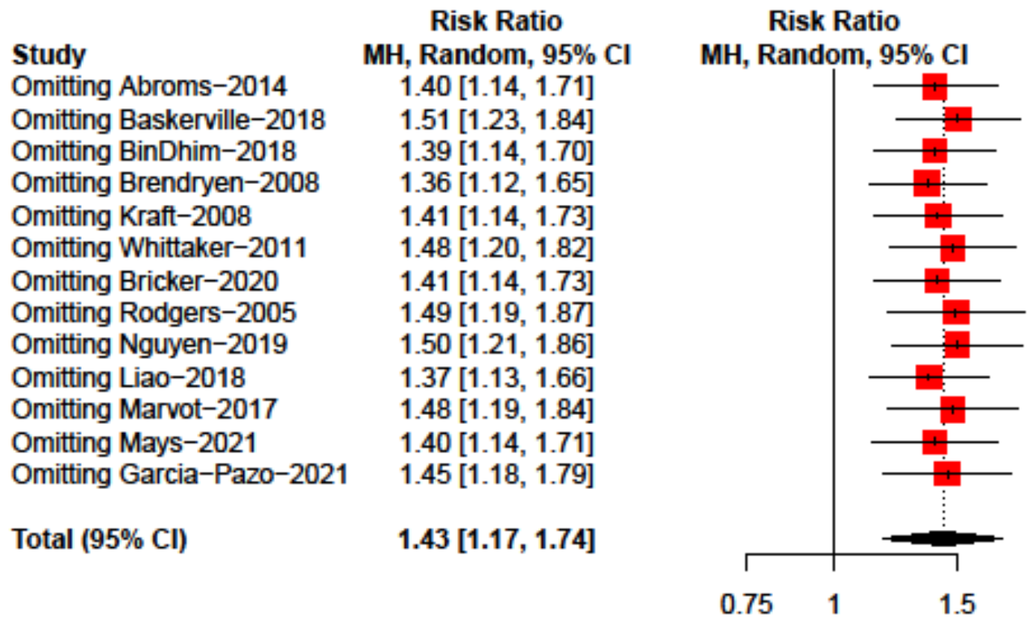


Supporting Information 5: sensitivity forest plot

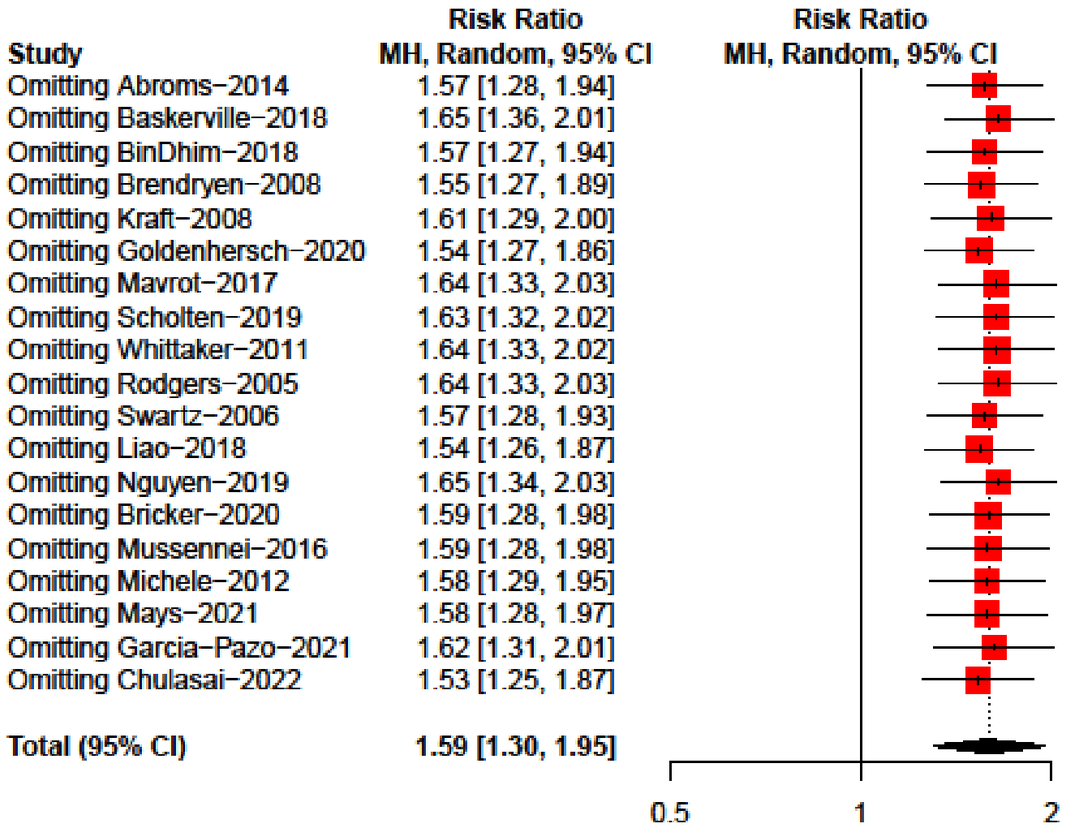
Supporting Information 5.1: sensitivity forest plot of 3-month abstinence



Supporting Information 5.2: sensitivity forest plot of 6-month abstinence



Supporting Information 5.3: Sensitivity forest plot of endpoint abstinence



Supporting Information 6 univariate regression results

Supporting Information 6.1 univariate regression of the TCS items, TCS categories and total score at 3-month

Theory coding scheme item (item number)	Item description	Studies where item coded as present, n (%)	Univariate		
			B	95%CI	p
Theory or model of behavior mentioned(I1)	Models or theories that specify relations among variables to explain or predict behavior are mentioned even if the intervention is not based on this theory	19(100)	/	/	/
Targeted construct mentioned as predictor of behavior (I2)	"Targeted" construct refers to a psychological construct that the study intervention is hypothesized to change	15(89)	0.0628	(-0.5317, 0.6573)	0.8359
Intervention based on single theory (I3)	The intervention is based on a single theory (rather than a combination of theories or theory and predictors)	2(11)	0.7665	(0.1981, 1.3349)	0.0082*
Theory or predictors used to select recipients for the intervention (I4)	Participants were screened or selected based on achieving a particular score or level on a theory-relevant construct or predictor	1(5)	0.3701	(-0.5705, 1.3107)	0.4406
Theory or predictors used to select or develop intervention techniques (I5)	The intervention is explicitly based on a theory or predictor or combination of theories and predictors	19(100)	/	/	/
Theory or predictors used to tailor intervention techniques to recipients (I6)	The intervention differs for different subgroups that vary on a psychological construct or predictor at baseline	1(5)	0.3701	(-0.5705, 1.3107)	0.4406
All intervention techniques are explicitly linked to at least one theory-relevant construct or predictor (I7)	Each intervention technique is explicitly linked to at least one theory-relevant construct or predictor	3(16)	0.8914	(0.3799, 1.4030)	0.0006*
At least one, but not all, of the intervention techniques are explicitly linked to at least one theory-relevant construct or predictor (I8)	At least one, but not all, of the intervention techniques are explicitly linked to at least one theory-relevant construct or predictor	18(95)	0.3210	(-0.4168, 1.0589)	0.3938
Group of techniques are linked to a group of constructs or predictors (I9)	A cluster of techniques is linked to a cluster of constructs or predictors	3(16)	0.1103	(-0.3006, 0.5212)	0.5988
All theory-relevant constructs or predictors are explicitly linked to at least one intervention technique (I10)	Every theoretical construct within a state theory, or every stated predictor, is linked to at least one intervention technique	1(5)	1.3819	(0.1034, 2.6604)	0.0341*
At least one, but not all, of the theory-relevant constructs or predictors are explicitly linked to at least one intervention technique (I11)	At least one, but not all, of the theoretical constructs within a stated theory or at least one, but not all, of the stated predictors (see I5) are linked to at least one intervention technique	17(89)	0.4330	(-0.1100, 0.9759)	0.1181
Theory-relevant constructs are measured: post intervention (I12a)	At least one construct of theory (or predictor) mentioned in relation to the intervention is measured post intervention	18(95)	0.3210	(-0.4168, 1.0589)	0.3938
Theory-relevant constructs are measured: post and pre intervention (I12b)	At least one construct of theory (or predictor) mentioned in relation to the intervention is measured pre and post intervention	10(53)	0.0913	(-0.2369, 0.4195)	0.5856
Changes in measured theory-relevant constructs or predictors (I13)	The intervention leads to significant change in at least one theory relevant construct or predictor (vs control group) in favor of the intervention	9(47)	0.2451	(-0.0564, 0.5466)	0.1111
Mediator predicts the dependent variable (I14a)	Mediator predicts dependent variable, or change in mediator leads to change in dependent variable	2(11)	0.1150	(-0.3947, 0.6246)	0.6584
Mediator predicts dependent variable, controlling for the independent variable (I14b)	Mediator predicts dependent variable when controlling for independent variable	2(11)	0.1150	(-0.3947, 0.6246)	0.6584
Intervention does not predict the dependent variable when controlling the independent variable (I14c)	Intervention does not predict dependent variable when controlling for mediator	0(0)	/	/	/
Mediated effect is statistically significant (I14d)	Mediated effect is statistically significant	2(11)	0.1150	(-0.3947, 0.6246)	0.6584

Results discussed in relation to theory (I15)	Results are discussed in terms of the theoretical basis of the intervention	8(42)	0.0636	(-0.2676, 0.3948)	0.7065
Appropriate support for theory (I16)	Support for the theory is based on appropriate mediation, or refutation of the theory is based on obtaining appropriate null effects (i.e., changing behavior without changing the theory-relevant constructs)	2(11)	0.1150	(-0.3947, 0.6246)	0.6584
Results used to refine theory: adding or removing constructs to the theory (I17a)	Authors attempt to refine the theory upon which the intervention was based by adding or removing constructs to the theory	0(0)	/	/	/
Results used to refine theory: specifying that the interrelationships between the theoretical constructs should be changed(I17b)	Authors attempt to refine the theory upon which the intervention was based by specifying that the interrelationships between the theoretical constructs should be changed and spelling out which relationships should be changed	0(0)	/	/	/

Theory coding scheme categories (category number)	**Category description**	**Items included**	**univariate**		
			B	**95%CI**	**p**
Reference to underpinning theory (C1)	Stated or suggested, rather than demonstrated theoretical base	1, 2, 3	0.3735	(-0.0231, 0.7700)	0.0649
Targeting of relevant theoretical constructs (C2)	Whether evidence was provided that a targeted theoretical construct predicted behavior, whether theory or predictors were explicitly used for designing the intervention, and the extent to which the intervention targets particular theory-relevant constructs	2, 5, 6, 7, 8, 9, 10, 11	0.2176	(0.0492, 0.3859)	0.0113*
Using theory to select recipients or tailor interventions (C3)	Whether theory was used to select participants likely to benefit from the intervention, or to tailor the intervention to the needs of a particular individual	4, 6	0.1851	(-0.2852, 0.6554)	0.4406
Measurement of constructs (C4)	Whether the relevant theory-based constructs or predictors have been measured	12a, 12b	0.1106	(-0.1689, 0.3900)	0.4380
Testing of theory: mediation effects (C5)	Whether theoretical constructs are measured, whether the intervention changes the theoretical constructs, and whether these changes explain the effect	12a, 12b, 13, 14a, 14b, 14c, 14d, 15, 16	0.0366	(-0.0398, 0.1129)	0.3477
Refining theory (C6)	Whether the results of evaluating theory-based interventions are used to refine theory	17a, 17b	/	/	/
Total use of theory	/	All items	0.0413	(-0.0066, 0.0892)	0.0908

* p-value < 0.05

Supporting Information 6.2 univariate regression of the TCS items, TCS categories and total score at 6-month

Theory coding scheme item (item number)	**Item description**	**Studies where item coded as present, n (%)**	**Univariate**		
			B	**95%CI**	**p**
Theory or model of behavior mentioned(I1)	Models or theories that specify relations among variables to explain or predict behavior are mentioned even if the intervention is not based on this theory	19(100)	/	/	/
Targeted construct mentioned as predictor of behavior (I2)	“Targeted” construct refers to a psychological construct that the study intervention is hypothesized to change	15(89)	0.3447	(-0.2362, 0.9255)	0.2449
Intervention based on single theory (I3)	The intervention is based on a single theory (rather than a combination of theories or theory and predictors)	2(11)	0.9416	(-0.0026, 1.8859)	0.0506
Theory or predictors used to select recipients for the intervention (I4)	Participants were screened or selected based on achieving a particular score or level on a theory-relevant construct or predictor	1(5)	/	/	/
Theory or predictors used to select or develop intervention techniques (I5)	The intervention is explicitly based on a theory or predictor or combination of theories and predictors	19(100)	/	/	/
Theory or predictors used to tailor intervention techniques to recipients (I6)	The intervention differs for different subgroups that vary on a psychological construct or predictor at baseline	1(5)	/	/	/
All intervention techniques are explicitly linked to at least	Each intervention technique is explicitly linked to at least one theory-relevant construct or	3(16)	0.3215	(-0.1659, 0.8089)	0.1961

one theory-relevant construct or predictor (I7)	predictor				
At least one, but not all, of the intervention techniques are explicitly linked to at least one theory-relevant construct or predictor (I8)	At least one, but not all, of the intervention techniques are explicitly linked to at least one theory-relevant construct or predictor	18(95)	0.1152	(-0.3711, 0.6015)	0.6425
Group of techniques are linked to a group of constructs or predictors (I9)	A cluster of techniques is linked to a cluster of constructs or predictors	3(16)	0.4094	(0.0134, 0.8054)	0.0427*
All theory-relevant constructs or predictors are explicitly linked to at least one intervention technique (I10)	Every theoretical construct within a state theory, or every stated predictor, is linked to at least one intervention technique	1(5)	/	/	/
At least one, but not all, of the theory-relevant constructs or predictors are explicitly linked to at least one intervention technique (I11)	At least one, but not all, of the theoretical constructs within a stated theory or at least one, but not all, of the stated predictors (see I5) are linked to at least one intervention technique	17(89)	0.4372	(-0.3287, 1.2031)	0.2632
Theory-relevant constructs are measured: post intervention (I12a)	At least one construct of theory (or predictor) mentioned in relation to the intervention is measured post intervention	18(95)	0.4372	(-0.3287, 1.2031)	0.2632
Theory-relevant constructs are measured: post and pre intervention (I12b)	At least one construct of theory (or predictor) mentioned in relation to the intervention is measured pre and post intervention	10(53)	0.0687	(-0.3625, 0.4999)	0.7548
Changes in measured theory-relevant constructs or predictors (I13)	The intervention leads to significant change in at least one theory relevant construct or predictor (vs control group) in favor of the intervention	9(47)	0.3521	(-0.0563, 0.7604)	0.0911
Mediator predicts the dependent variable (I14a)	Mediator predicts dependent variable, or change in mediator leads to change in dependent variable	2(11)	0.5009	(-0.0510, 1.0528)	0.0753
Mediator predicts dependent variable, controlling for the independent variable (I14b)	Mediator predicts dependent variable when controlling for independent variable	2(11)	0.5009	(-0.0510, 1.0528)	0.0753
Intervention does not predict the dependent variable when controlling the independent variable (I14c)	Intervention does not predict dependent variable when controlling for mediator	0(0)	/	/	/
Mediated effect is statistically significant (I14d)	Mediated effect is statistically significant	2(11)	0.5009	(-0.0510, 1.0528)	0.0753
Results discussed in relation to theory (I15)	Results are discussed in terms of the theoretical basis of the intervention	8(42)	0.0845	(-0.3097, 0.4787)	0.6745
Appropriate support for theory (I16)	Support for the theory is based on appropriate mediation, or refutation of the theory is based on obtaining appropriate null effects (i.e., changing behavior without changing the theory-relevant constructs)	2(11)	0.5009	(-0.0510, 1.0528)	0.0753
Results used to refine theory: adding or removing constructs to the theory (I17a)	Authors attempt to refine the theory upon which the intervention was based by adding or removing constructs to the theory	0(0)	/	/	/
Results used to refine theory: specifying that the interrelationships between the theoretical constructs should be changed(I17b)	Authors attempt to refine the theory upon which the intervention was based by specifying that the interrelationships between the theoretical constructs should be changed and spelling out which relationships should be changed	0(0)	/	/	/

Theory coding scheme categories (category number)	**Category description**	**Items included**	**univariate**		
			B	**95%CI**	**p**
Reference to underpinning theory (C1)	Stated or suggested, rather than demonstrated theoretical base	1, 2, 3	0.1113	(-0.0806, 0.3031)	0.2556
Targeting of relevant theoretical constructs (C2)	Whether evidence was provided that a targeted theoretical construct predicted behavior, whether theory or predictors were explicitly used for designing the intervention, and the extent to which the intervention targets particular theory-relevant constructs	2, 5, 6, 7, 8, 9, 10, 11	0.0540	(-0.0624, 0.1704)	0.3634
Using theory to select recipients or tailor interventions (C3)	Whether theory was used to select participants likely to benefit from the intervention, or to tailor the intervention to the needs of a particular individual	4, 6	0.0418	(-0.2365, 0.3201)	0.7685
Measurement of constructs (C4)	Whether the relevant theory-based constructs or predictors have been measured	12a, 12b	0.0751	(-0.1382, 0.2884)	0.4904
Testing of theory: mediation effects (C5)	Whether theoretical constructs are measured, whether the intervention changes the theoretical	12a, 12b, 13, 14a, 14b,	0.0660	(-0.0112, 0.1432)	0.0938

	constructs, and whether these changes explain the effect	14c, 14d, 15, 16			
Refining theory (C6)	Whether the results of evaluating theory-based interventions are used to refine theory	17a, 17b	/	/	/
Total use of theory	/	All items	0.0621	(0.0128, 0.1115)	0.0136*

* p-value < 0.05

Supporting Information 7： description of each item and category of the TCS

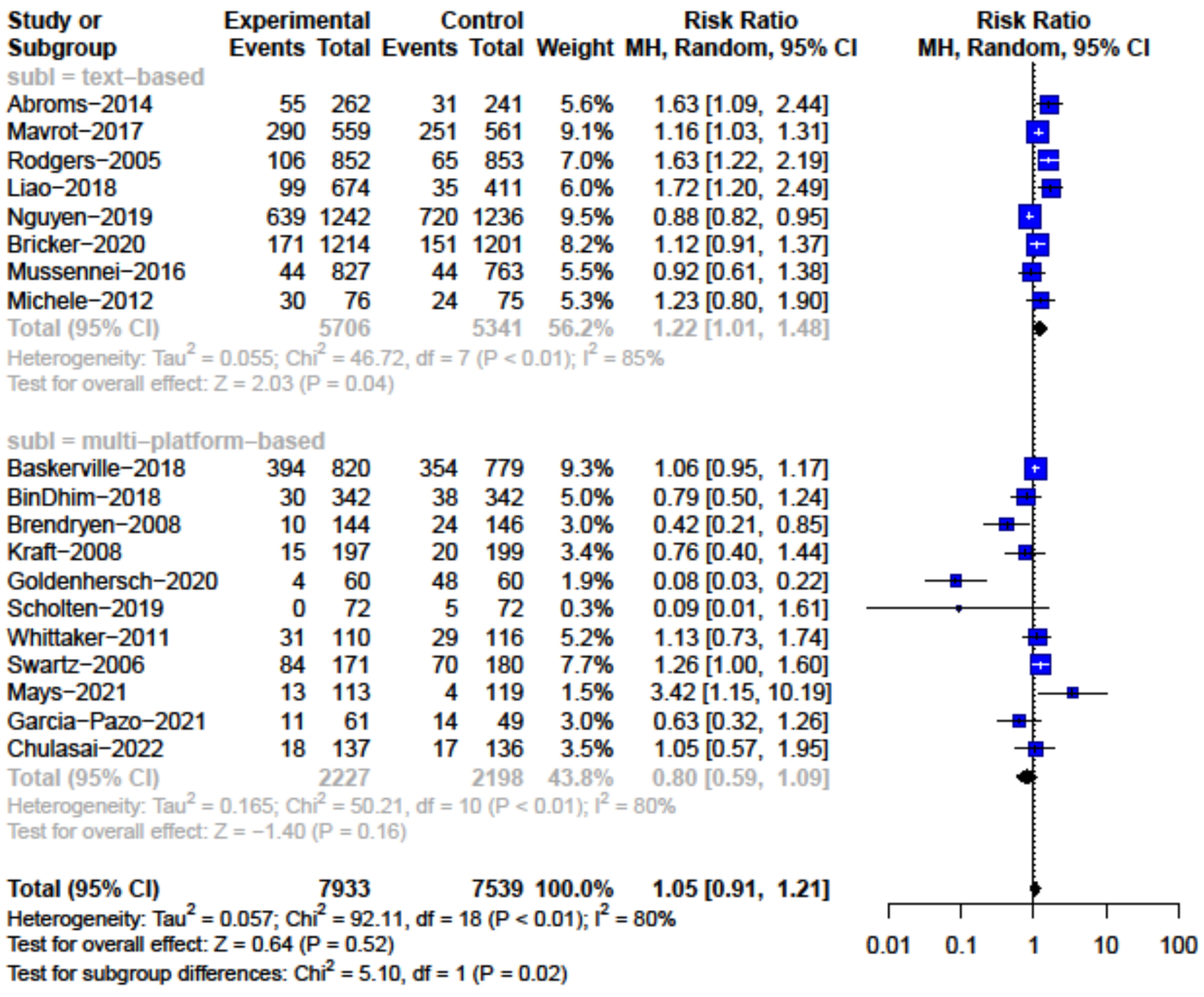
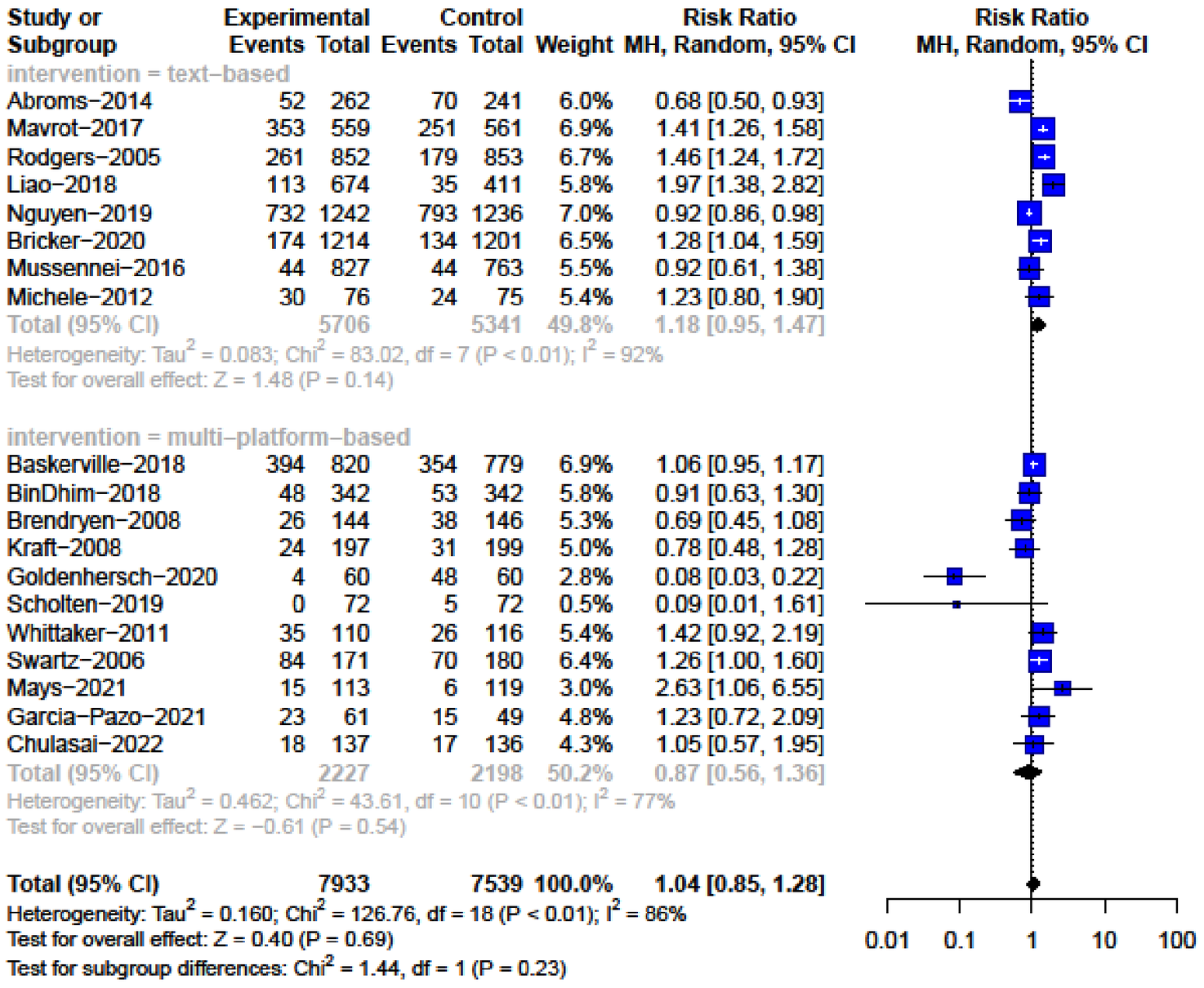
Theory coding scheme item (item number)	Item description
Theory or model of behavior mentioned(I1)	Models or theories that specify relations among variables to explain or predict behavior are mentioned even if the intervention is not based on this theory
Targeted construct mentioned as predictor of behavior (I2)	“Targeted” construct refers to a psychological construct that the study intervention is hypothesized to change
Intervention based on single theory (I3)	The intervention is based on a single theory (rather than a combination of theories or theory and predictors)
Theory or predictors used to select recipients for the intervention (I4)	Participants were screened or selected based on achieving a particular score or level on a theory-relevant construct or predictor
Theory or predictors used to select or develop intervention techniques (I5)	The intervention is explicitly based on a theory or predictor or combination of theories and predictors
Theory or predictors used to tailor intervention techniques to recipients (I6)	The intervention differs for different subgroups that vary on a psychological construct or predictor at baseline
All intervention techniques are explicitly linked to at least one theory-relevant construct or predictor (I7)	Each intervention technique is explicitly linked to at least one theory-relevant construct or predictor
At least one, but not all, of the intervention techniques are explicitly linked to at least one theory-relevant construct or predictor (I8)	At least one, but not all, of the intervention techniques are explicitly linked to at least one theory-relevant construct or predictor
Group of techniques are linked to a group of constructs or predictors (I9)	A cluster of techniques is linked to a cluster of constructs or predictors
All theory-relevant constructs or predictors are explicitly linked to at least one intervention technique (I10)	Every theoretical construct within a state theory, or every stated predictor, is linked to at least one intervention technique
At least one, but not all, of the theory-relevant constructs or predictors are explicitly linked to at least one intervention technique (I11)	At least one, but not all, of the theoretical constructs within a stated theory or at least one, but not all, of the stated predictors (see I5) are linked to at least one intervention technique
Theory-relevant constructs are measured: post intervention (I12a)	At least one construct of theory (or predictor) mentioned in relation to the intervention is measured post intervention
Theory-relevant constructs are measured: post and pre intervention (I12b)	At least one construct of theory (or predictor) mentioned in relation to the intervention is measured pre and post intervention
Changes in measured theory-relevant constructs or predictors (I13)	The intervention leads to significant change in at least one theory relevant construct or predictor (vs control group) in favor of the intervention
Mediator predicts the dependent variable (I14a)	Mediator predicts dependent variable, or change in mediator leads to change in dependent variable
Mediator predicts dependent variable, controlling for the independent variable (I14b)	Mediator predicts dependent variable when controlling for independent variable
Intervention does not predict the dependent variable when controlling the independent variable (I14c)	Intervention does not predict dependent variable when controlling for mediator
Mediated effect is statistically significant (I14d)	Mediated effect is statistically significant
Results discussed in relation to theory (I15)	Results are discussed in terms of the theoretical basis of the intervention
Appropriate support for theory (I16)	Support for the theory is based on appropriate mediation, or refutation of the theory is based on obtaining appropriate null effects (i.e., changing behavior without changing the theory-relevant constructs)
Results used to refine theory: adding or removing constructs to the	Authors attempt to refine the theory upon which the intervention was based by adding or removing constructs to the

theory (I17a)	theory
Results used to refine theory: specifying that the interrelationships between the theoretical constructs should be changed(I17b)	Authors attempt to refine the theory upon which the intervention was based by specifying that the interrelationships between the theoretical constructs should be changed and spelling out which relationships should be changed

Theory coding scheme categories (category number)	**Category description**
Reference to underpinning theory (C1)	Stated or suggested, rather than demonstrated theoretical base
Targeting of relevant theoretical constructs (C2)	Whether evidence was provided that a targeted theoretical construct predicted behavior, whether theory or predictors were explicitly used for designing the intervention, and the extent to which the intervention targets particular theory-relevant constructs
Using theory to select recipients or tailor interventions (C3)	Whether theory was used to select participants likely to benefit from the intervention, or to tailor the intervention to the needs of a particular individual
Measurement of constructs (C4)	Whether the relevant theory-based constructs or predictors have been measured
Testing of theory: mediation effects (C5)	Whether theoretical constructs are measured, whether the intervention changes the theoretical constructs, and whether these changes explain the effect
Refining theory (C6)	Whether the results of evaluating theory-based interventions are used to refine theory
Total use of theory	/

Supporting Information 8

Supporting Information 8.1 forest plot of lost to follow-up at 3-month (subgroups by intervention type)



Supporting Information 8.2 forest plot of lost to follow-up at endpoint (subgroups by intervention type)